

Alex Lenh

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EDUCATION

Bachelor of Science, Computer Science

Aug 2021 - May 2025

San Diego State University, San Diego, CA

GPA: 3.6/4.0

Relevant Coursework: Data Structures & Algorithms, Java OOP, Software Systems, Database Theory and Implementation, Computer Architecture, Operating Systems, Unix System Administrator, Artificial Intelligence, Machine Learning, C++ Programming

TECHNICAL SKILLS

Languages: Javascript/Typescript, C, C++, Java, Python, SQL

Frameworks & Libraries: React, Express, jQuery, Bootstrap, EJS, Tailwind, Spring Boot

Databases: PostgreSQL, SQLite, Sequelize ORM

Developer Tools: Bash, Git, Vite, VSCode, Postman, Node.js, Chrome DevTools, Docker, Linux, Unix, VMWare

PROJECTS

PityPlanner - Fullstack Web App | React, Typescript, SpringBoot, Postgres, Docker | Github Apr 2026 - Present

- Engineered a gacha-resource manager **aggregating user data** to project and calculate resource requirements, helping users optimize in-game resource management
- Designed and implemented a layered **REST API** app using **Spring Boot** and **React** to track gacha game resources, implementing a **PostgreSQL** schema to manage relationships for 5+ core entities
- Integrated Google **OAuth2** and **Spring Security**, implementing **stateless JWT authentication** to secure **30+ API endpoints** and token-based API access for all requests

Gacha Probability Engine | C++, CMake | Github

Mar 2026 - Apr 2026

- Built a Monte Carlo simulation engine in C++ with thread-safe task distribution, splitting runtime batches to parallelize **1M+ simulations**, **reducing elapsed runtime by ~8x**
- Developed a statistical **aggregation pipeline** over 1M+ simulation results, applying a **O(n log n) sort-once** with a result cache to optimize repeated queries in O(1) after the initial computation across merged thread results
- Implemented **custom binary serialization with RAII file guards** to compare and contrast simulation results across banner configs, with a histogram renderer that outputs ASCII distribution plots

Lock In - Chrome Extension | Typescript, Javascript, React | Github

Jan 2026 - Mar 2026

- Architected background scripts using **6+ Chrome APIs** to fetch **real-time user data** via **30 second polling intervals** to ensure data accuracy across restricted environments
- Developed an analytics frontend with **React** and **TypeScript** to provide **real-time data visualization** through **custom weighted algorithms** to calculate dynamic user-friendly metrics
- Optimized data synchronization performance by implementing a **100ms debounce** on rapid API events and **parallel async operations** to **reduce race conditions** and redundant storage writes calls

UNIX System Administration | VMWare, Bash, Azure, Linux, Unix

Jan 2025 - May 2025

- Built a **full-service virtual infrastructure** across Linux/Unix operating systems, including DNS, LDAP, mail, web, file, and print servers using FreeBSD, OpenBSD, Linux (Rocky/Ubuntu), Solaris, AIX, and Azure
- Automated SSH key generation and secure key distribution via Bash, **reducing manual key deployment time** across multiple Linux virtual machines
- Deployed and **managed a private and public cloud environment** with automated backups, P2V conversion, certificate management, and agent-forwarded secure access

Face Detection Model | Python, TensorFlow, ML, Data Training | Github

Mar 2024 - Apr 2024

- Led a team of five in developing a real-time AI facial recognition system using Python and TensorFlow, achieving **95% accuracy in diverse conditions**
- Managed **data collection and validation processes**, enhancing image processing capabilities for security and user authentication applications